1. HX-30HM Bus Servo User Manual



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- 1.1 Product Instruction
  - 1.1.1 Introduction
  - 1.1.2 Working Principle
  - 1.1.3 Servo Horn Installation & Pin Instruction
  - 1.1.4 Packing List
- 1.2 Parameter
- 1.3 Notice
- 1.4 Servo Testing Operations
  - 1.4.1 Magnetic Encoder Servo Debugging Board Introduction
  - 1.4.2 Hardware Connection
  - 1.4.3 PC Software Testing Description

1.1 Product Instruction

1.1.1 Introduction

HX-30HM Magnetic Encoder bus servo integrates servo drive, motor and bus servo signals. It is controlled by serial commands with a baud rate of 115200. Corresponding commands can be sent to the servo based on the communication protocol provided by Hiwonder to control its rotation or read its information. Three operation modes are available: Position Mode, Velocity (Closed-Loop) mode, and PWM (Open-Loop) Mode.

The interfaces of the servo are half duplex UART asynchronous serial interfaces. Each servo has three interfaces, allowing them to be daisy-chained. Theoretically, up to 253 bus servos can be connected in series. This servo is widely used in the joint design of various bionic.

1.1.2 Working Principle

A servo typically consists of a housing, a set of reduction gears, a motor, a potentiometer, and a control circuit.

Because the asynchronous serial bus's communication method involves data lines in series and parallel in the circuit, the controller must send corresponding instructions to the servo to achieve control. In serial communication, each transmission must be packaged and parsed according to a specific format.

The data frame format usually includes the frame header, ID number, data length, instruction, parameter and checksum field, wherein the frame header, that is, the start bit, each identification represents the start of new data transmission.

The ID number is used to distinguish different devices. The data length indicates the length of the data to be sent, where the data length plus 3 equals the length of the entire instruction packet, from the frame header to the checksum. The instruction field is used for various servo commands, such as position and speed control. The checksum is used to verify data integrity.

When the bus servo engages in serial communication, it first sends a request to the controller along with the corresponding parameters, such as position. The command packet includes the target servo's ID, and only the servo with the matching ID will execute the command, encapsulating the execution result and related status information in feedback information returned to the controller. The controller parses the command and performs the corresponding action.

This characteristic of bus servos enables the coordinated control and data exchange of multiple servos, greatly enhancing the system's scalability and flexibility, playing a crucial role in the components of robots.

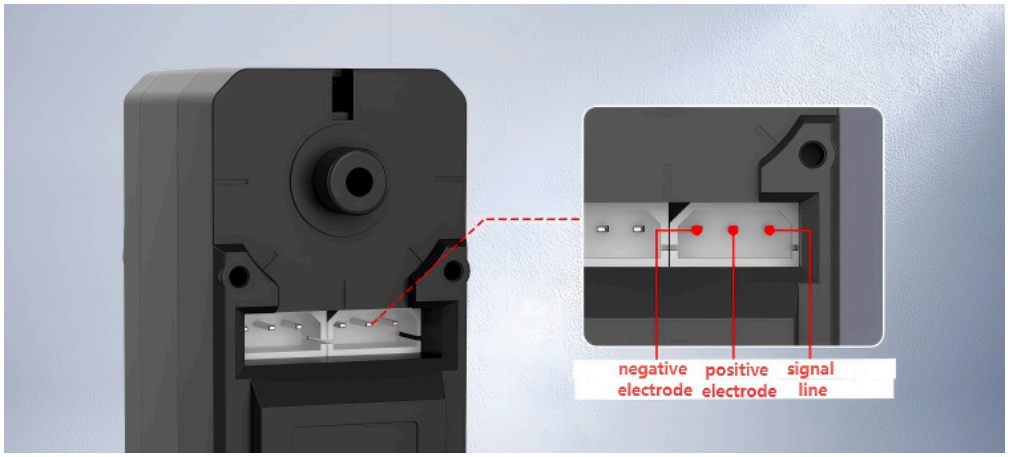
Regarding the Hiwonder Magnetic Encoder Bus Servo Communication Protocol Document, please refer to **Hiwonder Magnetic Encoder Bus Servo Communication Protocol Document\ 3 Servo Manual & Diagram**.

1.1.3 Servo Horn Installation & Pin Instruction

1. For the servo horn installation, please refer to the following picture: Taking the HX-30HM servo as an example, align and install the servo horn to the red cross direction shown in the image below when the servo is in its central position. If the horn cannot be installed in a cross orientation at neutral, refer to **1.4.3.2 Servo Information Reading and Setting** and use the position calibration function in the motion control settings to adjust the servo deviation.



2. The port distribution and instruction refer to the following picture and table.

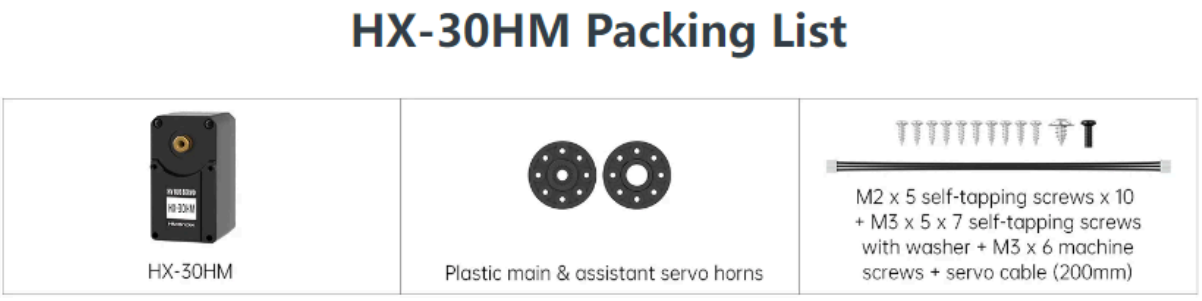


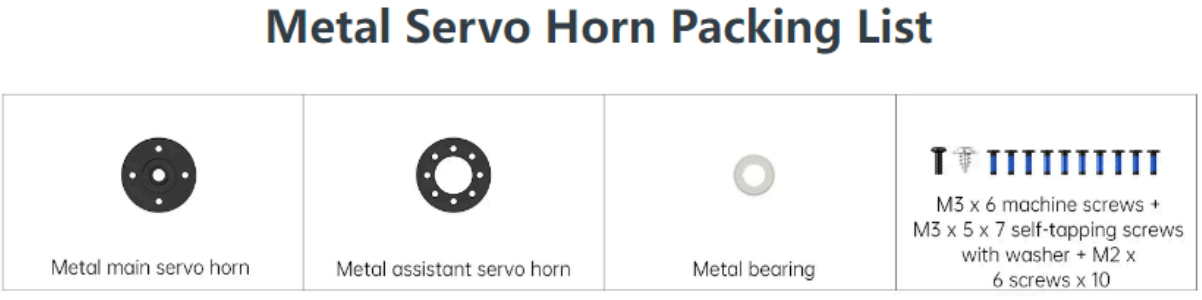
Pin (from left to right)	Instruction
GND	Power ground
VIN	Power input
SIG	Signal terminal for half-duplex UART asynchronous serial interface

> **Note**
>
> **When the servo and the microcontroller are not using the same power supply, ensure that the two power supplies share a common ground.**

1.1.4 Packing List

The servo and its accessories included in the shipment are shown in the image below:





1.2 Parameter

Product	**HX-30HM Magnetic Encoder Linear Bus Servo**	**Control method**	**UART serial command**
Brand	Hiwonder	Communication baud rate	1000000
Weight	52g	Memory	Servo settings are automatically saved when power off
Dimensions	45.2x24.7x35mm	Servo ID	0~253 for user setting, ID 1 is default
Working voltage	Rated 11.1 V, range 9–12.6 V	Readback function	Support angle readback
Rotation speed	0.19sec/60° 11.1V（HX-30HM）	Protection	Stall protection and over-temperature protection
Servo torque	30KG·cm 11.1V(HX-30HM)	Parameter feedback	Temperature, voltage and position
Rotation range	0° ~ 360°	Working mode	Command Mode and Step Mode
No-load current	100mA	Gear type	Metal gear
Stall current	3A	Servo wire length	20cm, optional wire lengths available
Servo precision	0.3°	Connector number	5264-3P
Control angle range	0-4095, corresponding to 0°~ 360°	Application	All kinds of bionic robot related joints

1.3 Notice

Please carefully read the following precautions before operations:

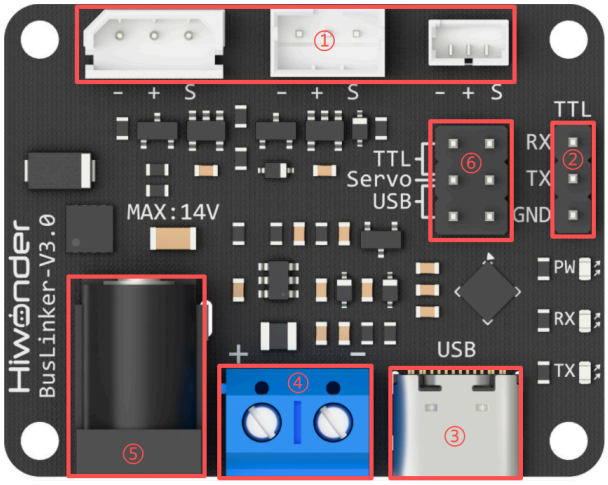
1. The operating voltage of this bus servo is 9-12.6V. Please select a stable power supply within this range. Do not use over-voltage as it will damage the servo. Conversely, insufficient voltage will prevent the servo from operating.
2. The default ID of the bus servo is 1. Before use, preset the servo ID according to your requirements.
3. Before setting the ID number for servo, connect each servo individually; otherwise, all servos will be set to the same ID. Connect servos in series after setting the IDs.

4. The bus servo is a high-precision accessory. After powering on, do not manually force the servo arm, servo horn, etc., to avoid internal damage to the servo.
5. The bus servo will heat up after prolonged operation. Allow the servo to cool down after running for a while to prevent overheating, which can affect performance and service life.
6. During continuous rotation, the load on the bus servo should not exceed its stall torque. It is recommended that the load be 1/5 to 1/3 of the stall torque.
   Stall torque refers to the maximum torque the servo can provide when it is unable to rotate, which means stalled or blocked. Due to energy loss during servo operation, control the load appropriately to avoid exceeding the servo's rated load. Excessive load will overwork the servo, increasing energy consumption and reducing efficiency.
7. The communication lines of the bus servo are susceptible to electromagnetic interference. To ensure stable and reliable communication, avoid placing the bus servo near high-power power lines, motor lines, or other potential sources of electromagnetic interference.
8. Set the angle control range and speed of the bus servo reasonably based on the actual project requirements to ensure safe operation. Avoid exceeding mechanical limits or moving too quickly to prevent accidental collisions or damage.

1.4 Servo Testing Operations

1.4.1 Magnetic Encoder Servo Debugging Board Introduction

The Magnetic Encoder Servo Type debugging board is a tool for debugging servos. With the provided ServoStudio, it can test servos and set servo parameters. Additionally, the debugging board can communicate via the serial port with a microcontroller to control the servos.



Index Instruction	Function Instruction
① 3pin Bus Servo Interface	Connecting and debugging Bus Servo
② Serial Port Pins	It communicates with other MCUs via the serial port
③ Type-c Interface	It communicates with PC software/hardware Controller
④ Power Terminal	Lithium Battery Power Connection
⑤ DC Interface	Adapter Power Connection

Index Instruction	Function Instruction
⑥ Communication Mode Select Pins	Switch Communication Mode: USB/TTL

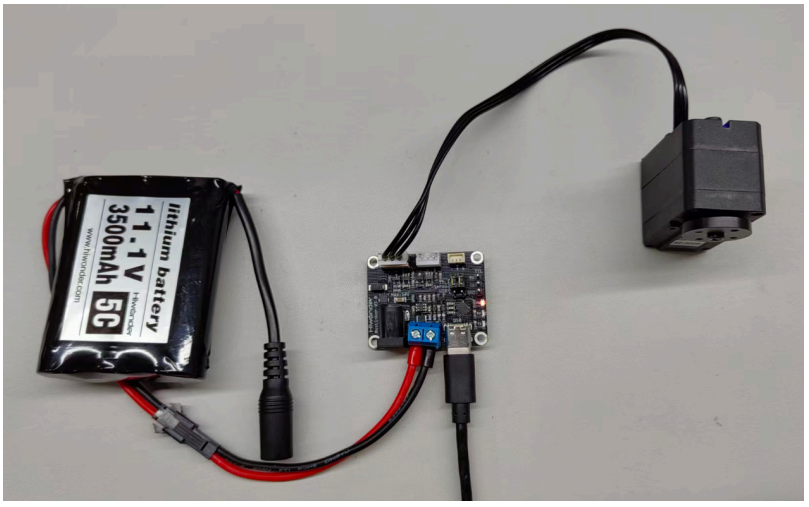
1.4.2 Hardware Connection

1. Connect the servo to the BusLinker V3.0 board via the servo cable to the 5264 3-pin bus servo port.
2. Connect the red wire to the “+”, the black wire to “-”, then connect USB cable.

> **Note**
>
> - **Ensure the battery polarity is correct when connecting the adapter cable. Do not reverse the positive and negative terminals.**
> - **Place the jumper cap on the Servo and USB pins.**

3. Connect the lithium battery to the battery connector cable. The cable is designed to prevent reverse insertion. Do not force it if it does not fit.

   If a lithium battery is not available, use a stable 12 V power supply.



1.4.3 PC Software Testing Description

1.4.3.1 Servo Status Description

The following three images show the three basic servo positions — central position, maximum angle, and minimum angle. For visual demonstration, a servo horn is inserted as an example, with the label placed on the right-side servo arm to indicate the initial position.



Central position (Adjustable): 180° servo angle, corresponding to the position of 2047.



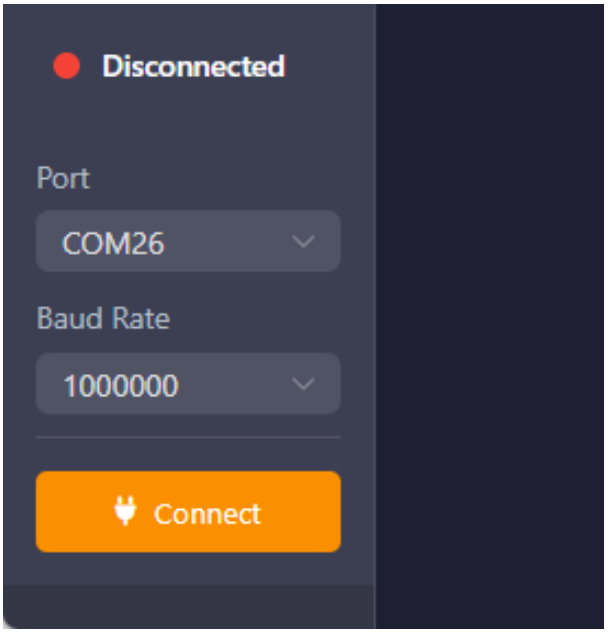
Maximum position per revolution: 360°servo angle, corresponding to the position of 4095



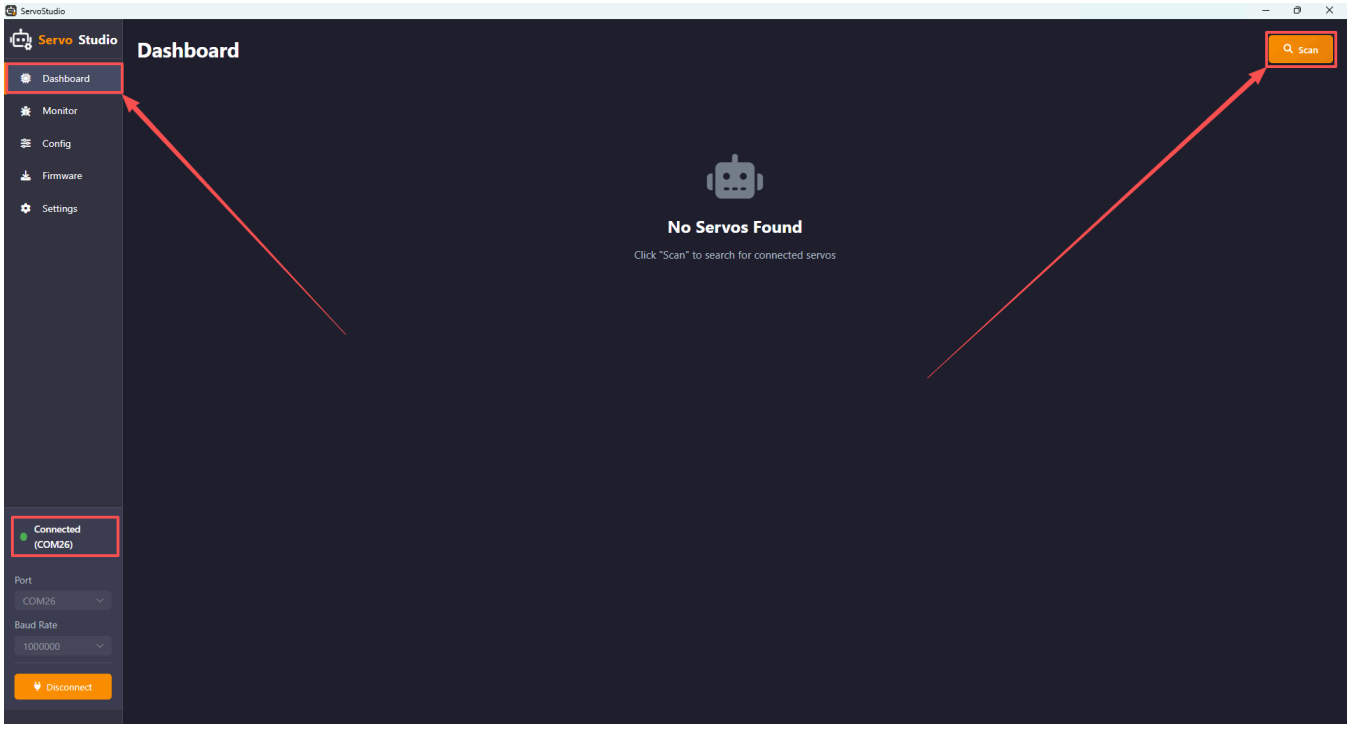
Minimum Position per revolution: 0° servo angle, corresponding to the position of 0.

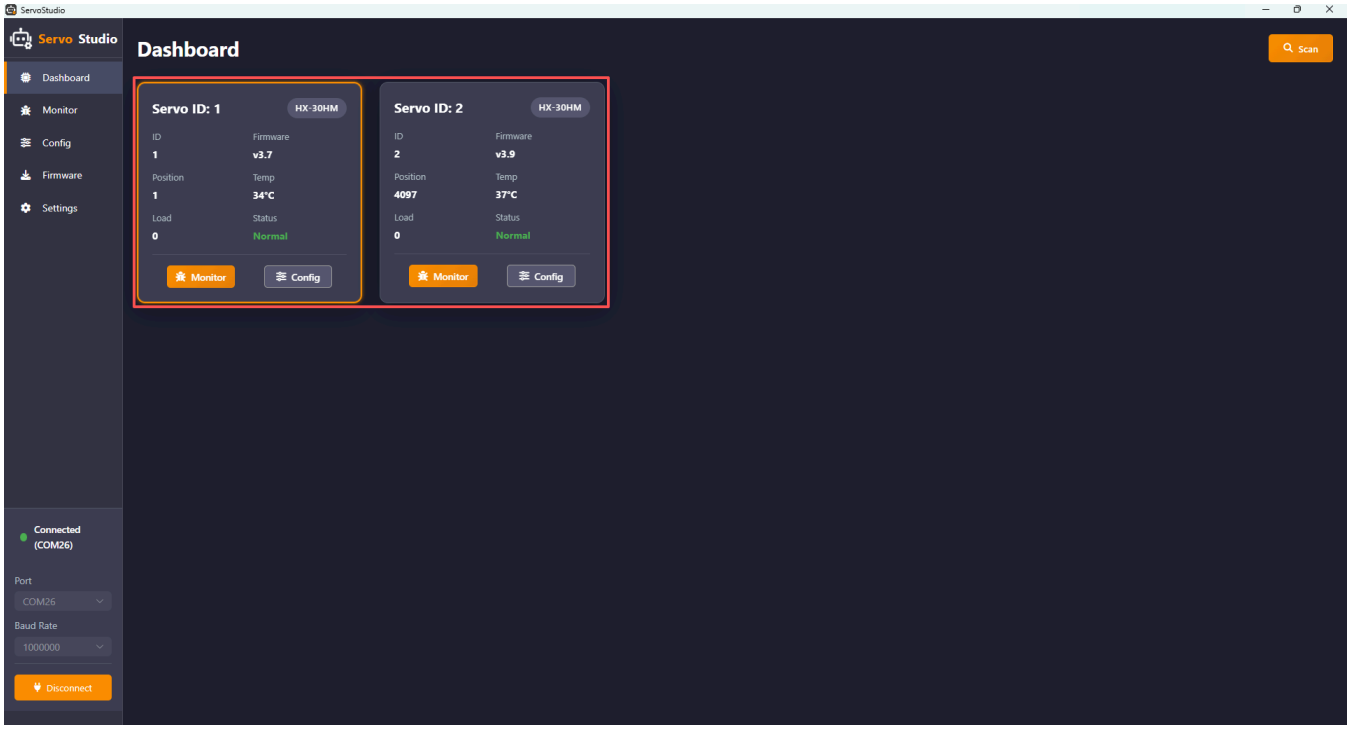
1.4.3.2 Reading and Configuring Servo Information

1. Open the BusLinker V3.0 board software located at **2 Softwares / Servo Debugging Board Software**. On the left side of the interface, select the serial port, then click the button to open the port. The default baud rate is 1,000,000, then click **Connect**.

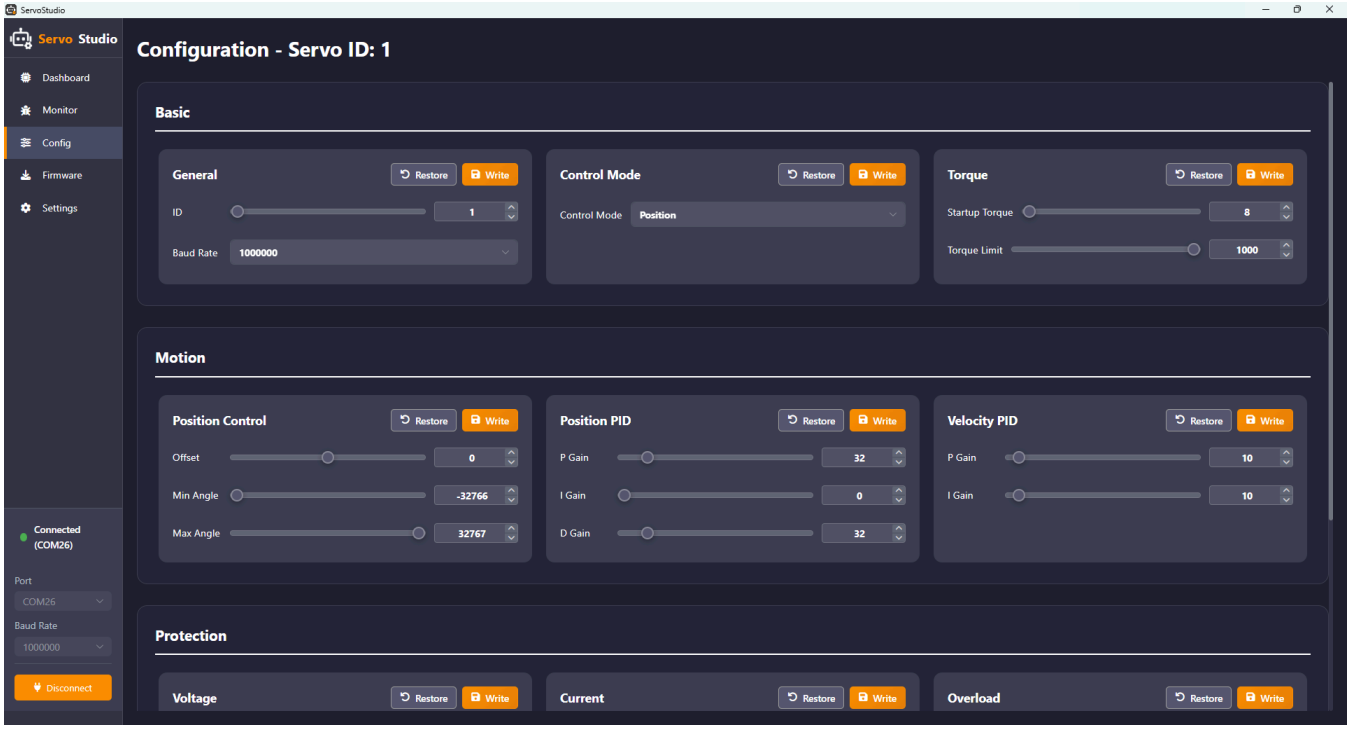


2. After the serial port is connected, click **Scan** in the Dashboard. The software will scan servos with IDs from 1 to 253. The detected servos will be displayed in the interface.



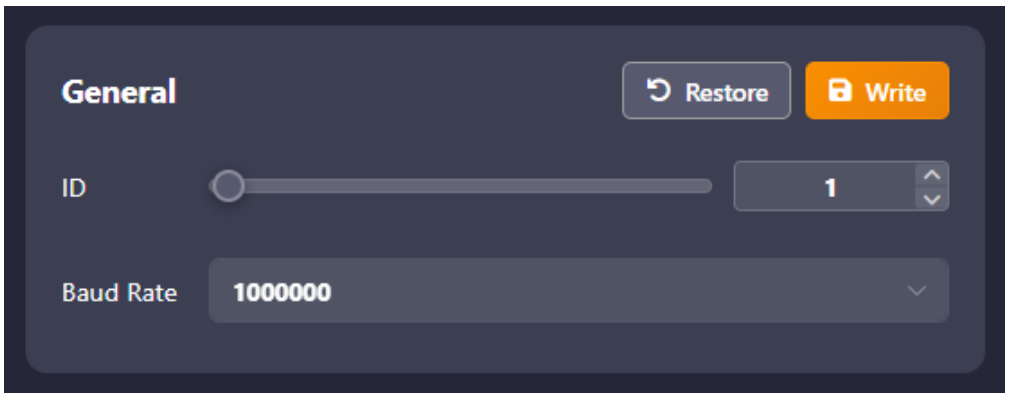


3. Click **Config** to enter the parameter configuration interface. Here, the servo parameters can be viewed and configured as needed.



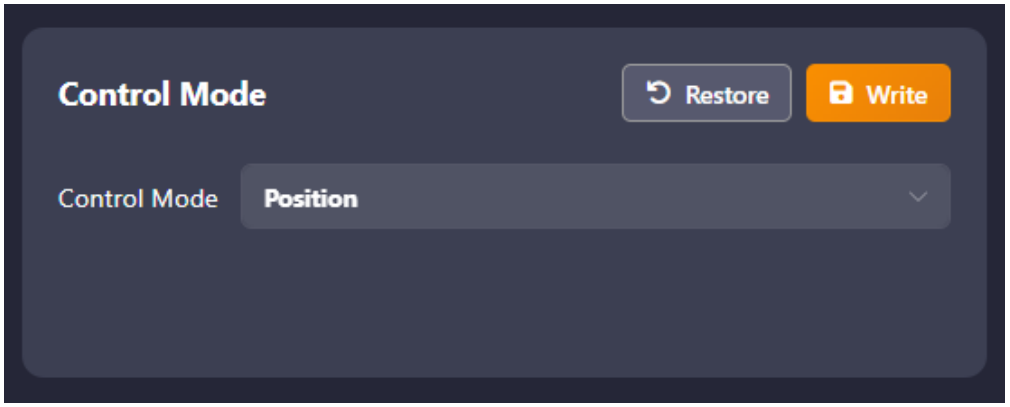
(1) Basic Configuration

① Basic Settings



In this module, the servo ID and the baud rate for communication between the PC software and the servo can be changed. Servo IDs range from 1 to 253, and eight baud rate options are available.

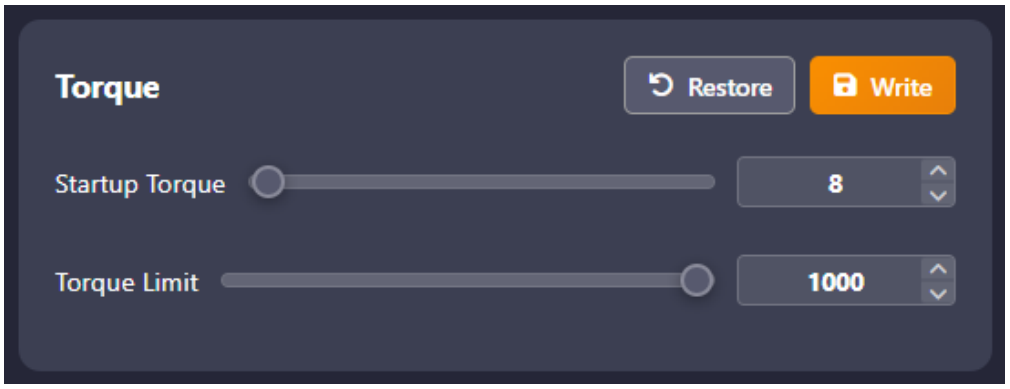
② Operation Mode



In this module, the servo can be set to three modes: Position, Velocity (Closed-Loop), and PWM (Open-Loop).

In Position mode, the servo operates in servo mode. In both Velocity (Closed-Loop) and PWM (Open-Loop) modes, the servo operates in Motor Mode.

③ Torque Control

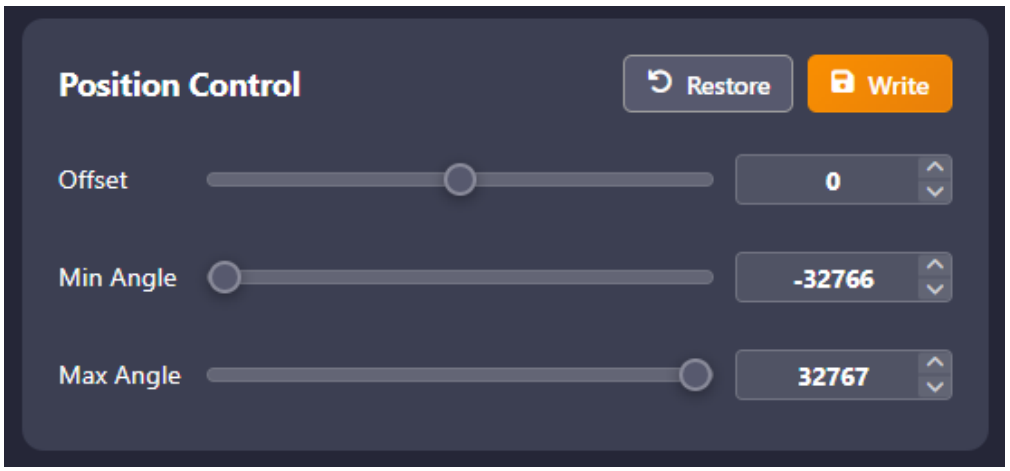


This control mode is used to set the servo's minimum starting torque and maximum output torque limits. Set 1000 = 100% × Stall Torque.

> **Note**
>
> **If the torque is set too low, the servo may fail to rotate.**

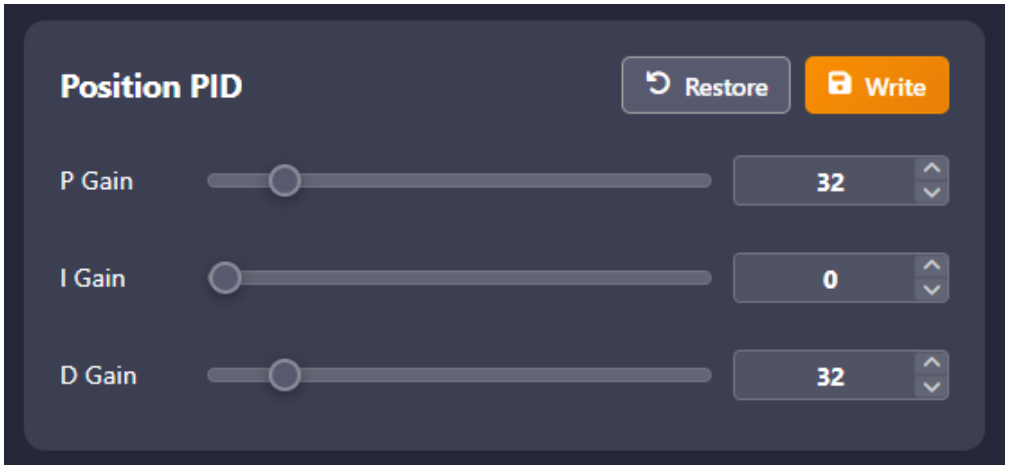
(2) Motion Control

① Position Control Parameters



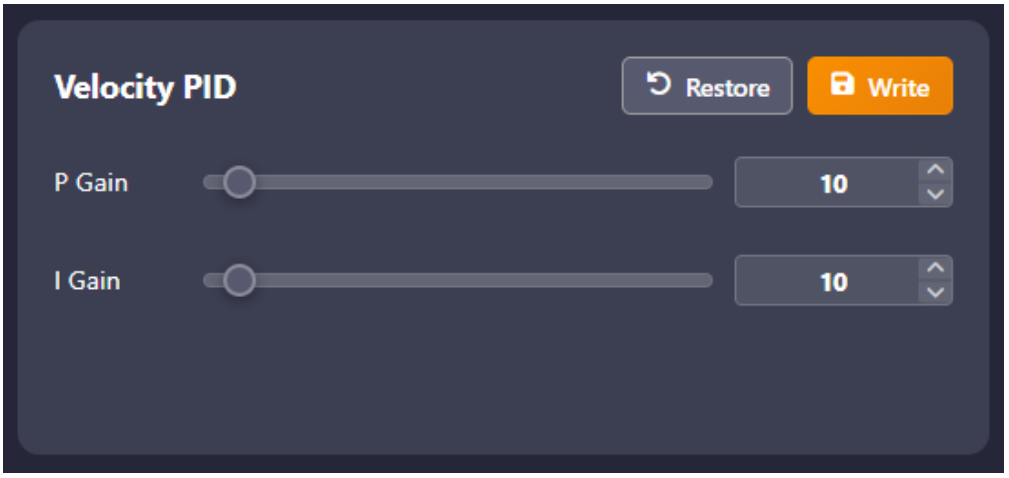
- Offset — Position correction used to calibrate servo deviation. The range is -2047 to 2047. If the servo horn cannot be installed in a cross posture due to deviation, adjust this parameter to correct the offset.
- Min/Max Angle — Minimum and maximum angle limits for restricting the servo rotation range. The range is -30719 to 30719. In absolute position mode, the maximum servo position range is -30719 to 30719. Position 0 to 4095 corresponds to 0° to 360°, so the maximum rotation range is approximately ±7.5 turns. The rotation range can be set within this range according to requirements.

② Position Closed-Loop Parameters



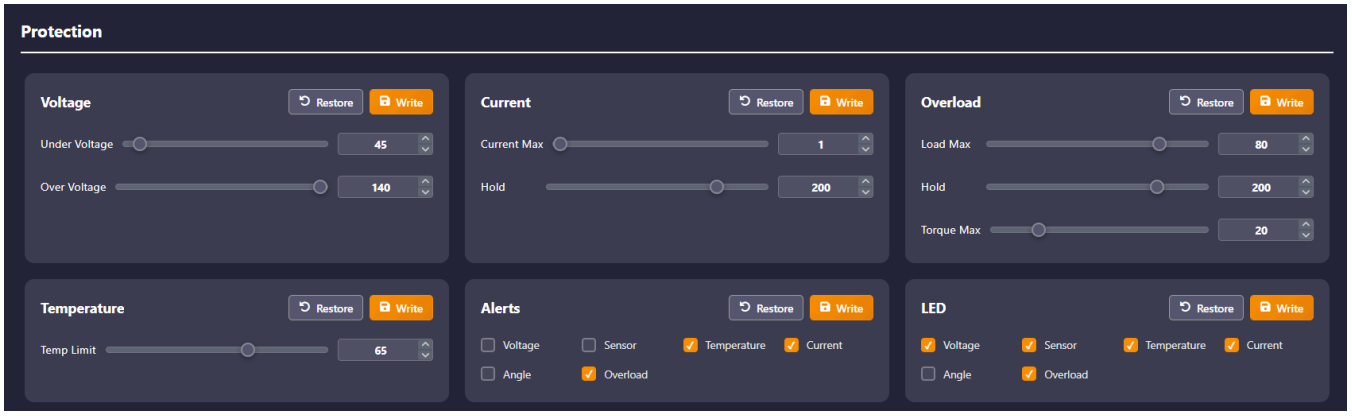
Three parameters are used to adjust the servo position error: **P** determines the influence of the current position error on the control signal. **I** reduces long-term static error and normally does not need adjustment. **D** reduces the rate of change of error, improving system stability. These parameters can be adjusted according to requirements to achieve the desired performance.

③ Speed Closed-Loop Parameters



Two parameters are used to adjust the servo speed error, ensuring motion at the desired speed and maintaining precise speed control. These parameters can be adjusted according to requirements to achieve the desired performance.

(3) Safety Protection

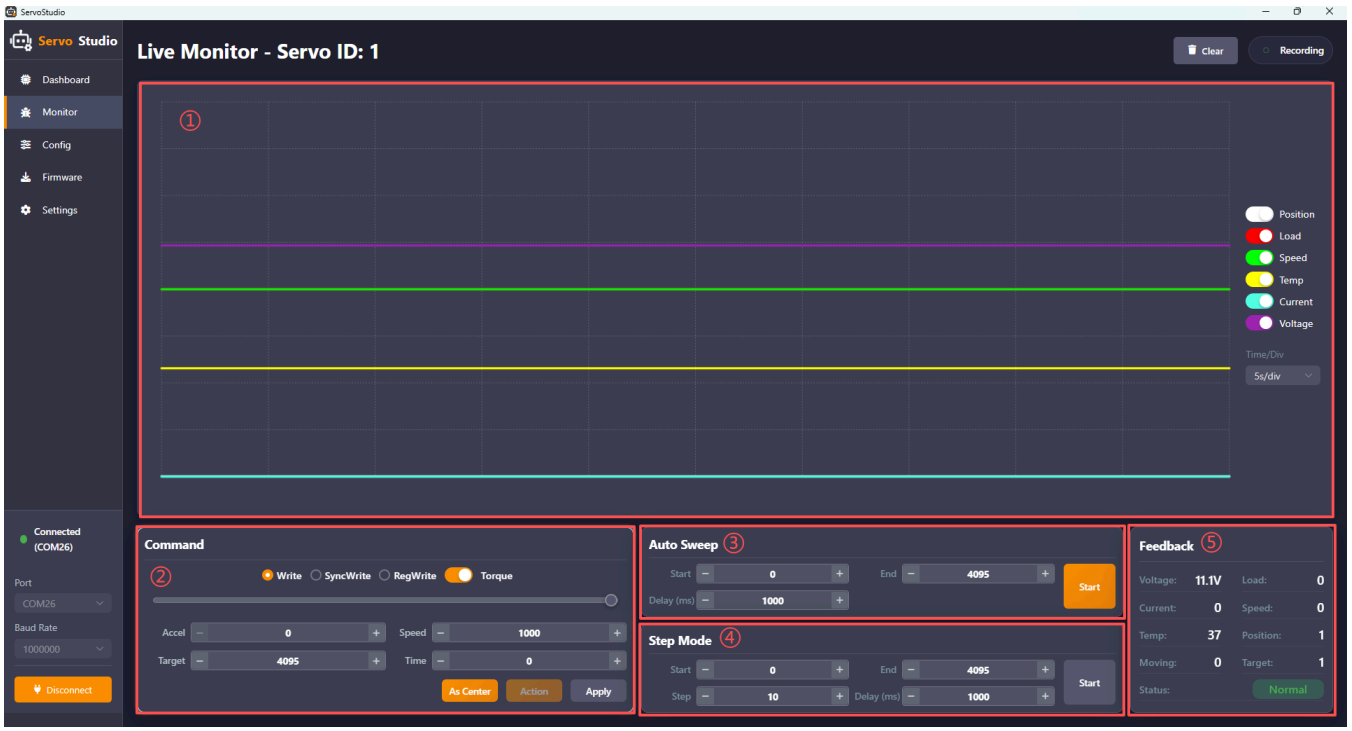


Thresholds for each module can be adjusted and the actions that trigger protection or alarms can be selected according to requirements.

- Voltage — Under Voltage and Over Voltage threshold, range 0–254, unit 0.1 V.
- Current — Current Max threshold, range 0–6000, unit mA. Hold is the overcurrent protection time threshold, with a range of 0–254, and a unit of 10 ms.
- Overload — Load Max, range 0–100, unit 1%, maximum corresponds to the maximum torque. Hold is the overload time threshold, with a range of 0–254, and a unit of 10 ms. Torque Max is the maximum output torque after triggering overload protection, with a range of 0–100, and a unit of 1%.
- Over-temperature protection threshold, range 0–100, unit °C.

1.4.3.3 Servo Operation Control

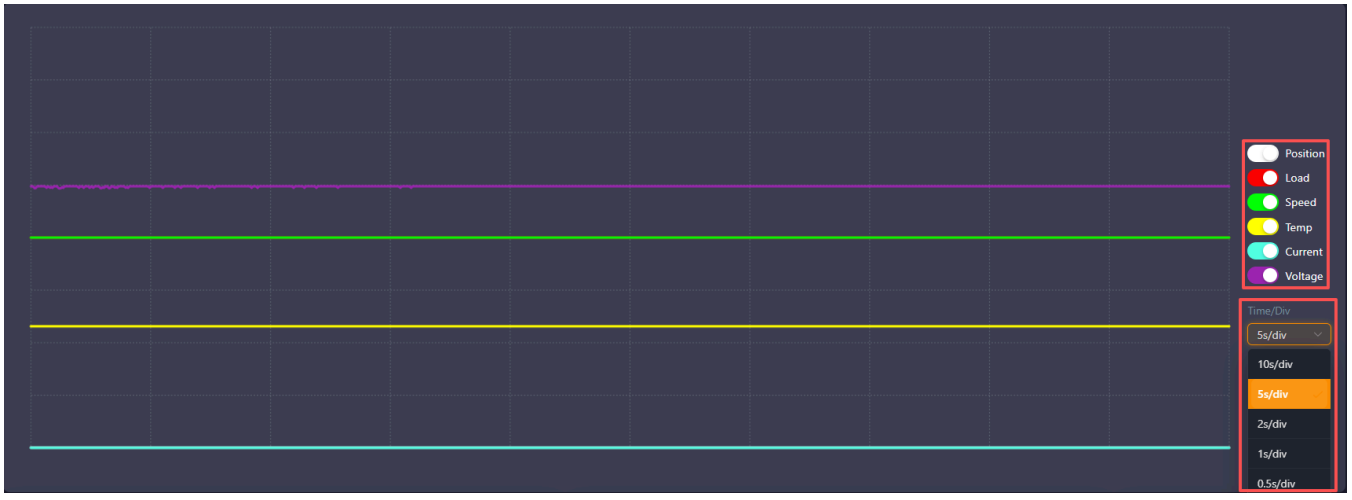
Click the **Monitor** option on the left to enter the servo debugging interface.



The interface consists of five areas:
① Servo operation parameter waveform area. ② Command area for servo operation. ③ Auto Sweep area. ④ Step Mode area. ⑤ Feedback area.

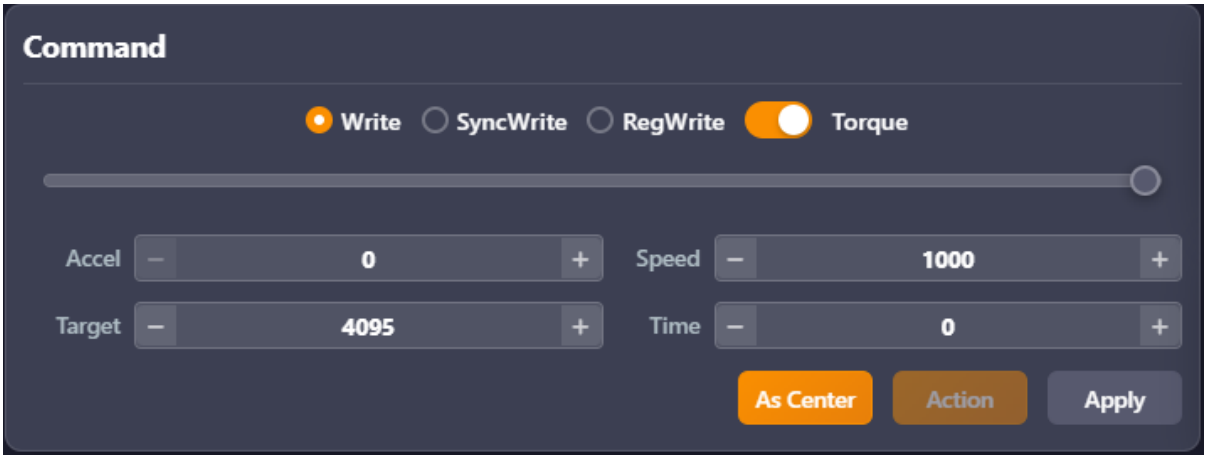
The five areas are described as follows:

① Servo Operation Parameter Waveform Area



This area provides a visual display of the changes in six servo parameters during operation. On the right, switches can be enabled to display the desired parameters. Below the parameter switches, a dropdown menu allows selection of the time per division for the waveform display.

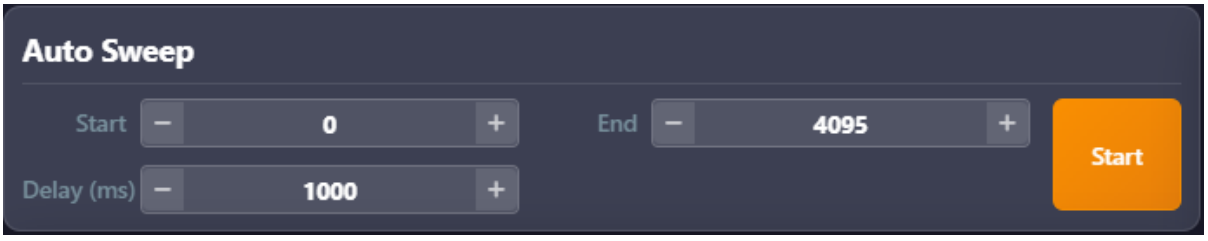
② Servo Operation Area



This area allows configuration of the three servo write modes: **Write**, **SyncWrite**, and **RegWrite**. Detailed information about these write modes can be found in the [Hiwonder Magnetic Encoder Bus Servo Communication Protocol Document]. The **Torque** switch controls whether the servo is powered. When the switch is turned off, the servo stops outputting torque. If a servo horn is installed, it can then be manually rotated. **The slider** in the middle can be used to directly control the servo position, with a range of 0–4095. Below the slider, the following parameters can be set for servo movement:
Accel — acceleration, range 0–254, unit 100 steps/s².
Speed — speed, range 0–3400, unit steps/s.
Target — target position, range -30719 to 30719, unit steps.
Time — time, range 0–1000. At the bottom, three buttons are provided:
As Center sets the current target position as the central position. **Action** triggers motion in asynchronous write mode. **Apply** sends parameters to the servo in normal write and synchronous write modes.

> ⓘ **Note**
>
> - **In Position mode, the Time parameter is ignored.**
> - **In Velocity (Closed-Loop) mode, only the Speed parameter is effective.**
> - **In PWM (Open-Loop) mode, only the Time parameter is effective.**

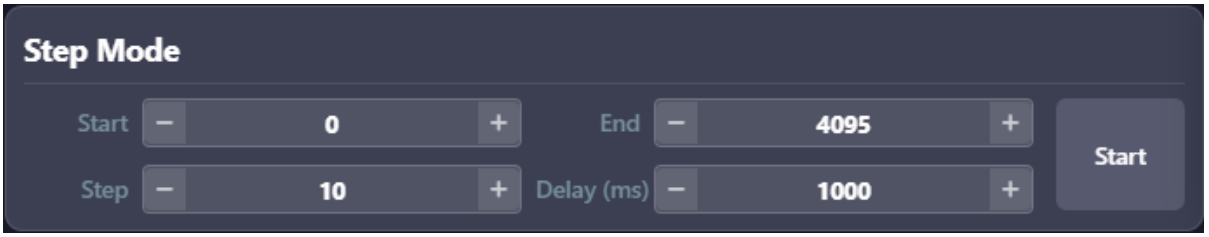
③ Auto Sweep Area:



This area allows setting the **Start** and **End** positions of the servo, with a range of -30719 to 30719 in steps. **Delay** specifies the interval time between sending the **Start** and **End** positions. **Speed** is set in the Command area for servo operation. Clicking **Start** causes the servo to move back and forth between the two positions.

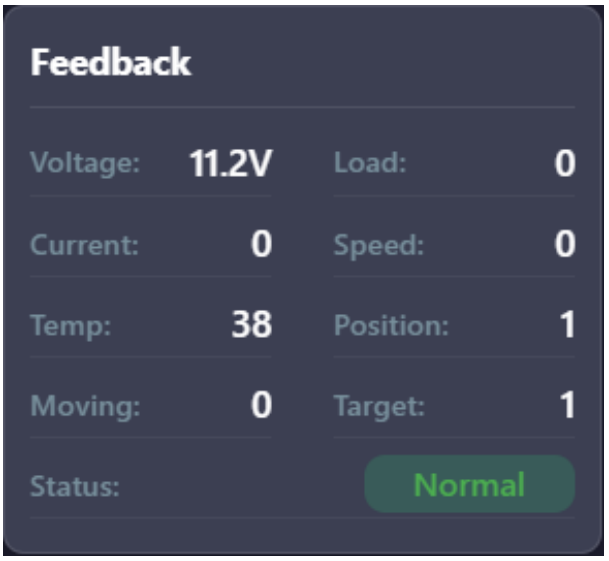
> ⓘ **Note**
>
> **If the delay is shorter than the time required to move between the two positions, the current motion will be interrupted.**

④ Auto Step Mode Area:



In this area, the **Start** and **End** positions of the servo can be set, with a range of -4095 to 4095 in steps. **Step** specifies the step size, with a range of 0–1000 in steps. **Delay** specifies the interval time between steps. Clicking **Start** causes the servo to move from the **Start** to the **End** position according to the specified **Step** and **Delay** values.

⑤ Feedback Area



This area provides a visual display of the servo status and all relevant parameters during operation.